

Esharaqa Afreen Jahid

765-694-5019 | ejahid@purdue.edu | [linkedin.com/in/esharaqa-jahid](https://www.linkedin.com/in/esharaqa-jahid) | [Portfolio](#)

EDUCATION

Purdue University

M.S. in Electrical and Computer Engineering

West Lafayette, IN

Jan 2026 – Dec 2026

Purdue University

B.S. in Computer Engineering; GPA: 3.43

West Lafayette, IN

Aug 2022 – May 2026

TECHNICAL SKILLS

Logic Design & Architecture: SystemVerilog, Verilog, VHDL, RISC-V Pipeline Design, Cache Coherence, AHB Bus
Verification Methodologies: UVM, SVA, Constrained Random Verification, Formal Verification, Failure Analysis
Embedded & Physical Systems: PCB Layout, I2C, SPI, UART, Firmware (Bare-metal C), USB
EDA Tools & Software: Cadence Innovus, Xilinx Vivado, ModelSim, C/C++, Python, Bash/Shell, Linux, Git, Kicad, LTspice

EXPERIENCE

IC Design Intern

Fusion Semiconductor

May 2025 – Aug 2025

Remote

- Designed an AHB-Lite subordinate interface with pipelined writes in SystemVerilog; resolved RAW hazards via forwarding logic to ensure logic design specification compliance prior to synthesis.
- Resolved post-synthesis timing failures from asynchronous flush logic by refactoring to a synchronous reset, successfully achieving timing closure targets and eliminating write-flush race conditions.
- Built UVM testbenches with SVA for formal verification of protocol compliance; performed rigorous failure analysis on complex corner cases that escaped directed testing.

Teaching Assistant

ECE 362 – Microprocessor Systems and Interfacing, Purdue University

Aug 2025 – Present

West Lafayette, IN

- Mentor students in bare-metal C on RP2350, debugging interrupt-driven firmware including ISR design, DMA configuration, and direct register manipulation.
- Validated RP2350-based course hardware during initial deployment; identified CMSIS-DAP debug probe firmware incompatibility and documented GPIO pin behavior requiring pull-down workarounds.

PROJECTS

Pipelined Multicore RISC-V Processor

West Lafayette, IN

- Implemented a dual-core RV32I processor with a 5-stage pipeline, private L1 caches, and MEM-to-EX forwarding in SystemVerilog; verified through testbenches and deployed to a Xilinx FPGA.
- Restructured combinational control logic for timing closure, improving operating clock frequency from 73 MHz to 96 MHz without modifying the underlying computer architecture.
- Analyzed CPI, cache hit rates, and critical path timing to perform processor characterization, validate specification compliance, and eliminate microarchitectural bottlenecks.

Atalla 0x01 – AI Hardware Accelerator (Purdue SoCET)

West Lafayette, IN

- Designing a BF16 exponential (e^x) unit for an open-source AI accelerator vector lane; architecting area, power, and latency trade-offs for embedded deployment.
- Implementing the exponential function via a 64-entry $\times$ 16-bit LUT with piecewise linear interpolation in SystemVerilog; developing UVM testbenches with full-sweep BF16 coverage to verify numerical accuracy.

Pixel Party – Embedded Product Development Capstone

West Lafayette, IN

- Designed a dual-processor architecture (STM32 + Raspberry Pi) with UART state synchronization; implemented USB HID firmware enabling plug-and-play gamepad compatibility with Linux hosts.
- Executed printed circuit board (PCB) physical design and layout, routing high-speed USB differential pairs and optimizing decoupling capacitor placement near IC power pins to minimize noise coupling.